EA-0019 - AQELYN Engineering Archive

AQELYN - EA-0019 Engineering Archive

IS-019 - AQELYN Security Data Lake & Telemetry Platform

Archive ID: EA-0019

Implementation Specification: IS-019

Component: AQELYN Security Data Lake & Telemetry Platform

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: COMPLETE

Repository Impact: No top-level repository structure changes

Breaking Changes: None

Engineering Phase: Phase 3

Predecessor Archives: EA-0001 through EA-0018

Next Specification: IS-020 - AQELYN AI Decision Intelligence Engine

Document Control

Field	Value
Document	Engineering Archive EA-0019
Specification	IS-019 - AQELYN Security Data Lake & Telemetry Platform
Publication Format	Markdown, PDF, HTML, ZIP
Source of Truth	MD/EA-0019.md
Archive Rule	Implementation Specification -> Engineering Archive -> Continue
Repository Rule	Fixed repository structure; no redesign
Completion State	IS-019 complete; EA-0019 generated

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

Engineering Archive	Implementation Specification	Component
EA-0001	IS-001	AQELYN Kernel
EA-0002	IS-002	Universal Object Model
EA-0003	IS-003	AQELYN Event Bus

| EA-0004 | IS-004 | AQELYN Evidence Engine |

| EA-0005 | IS-005 | AQELYN Knowledge Graph |

| EA-0006 | IS-006 | AQELYN Trust Engine |

| EA-0007 | IS-007 | AQELYN Mission Engine |

| EA-0008 | IS-008 | AQELYN Workflow Engine |

| EA-0009 | IS-009 | AQELYN Policy Engine |

| EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |

| EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |

| EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |

| EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |

| EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |

| EA-0015 | IS-015 | AQELYN Security Operations (SOC) Engine |

| EA-0016 | IS-016 | AQELYN Digital Forensics Engine |

| EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |

| EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |

| EA-0019 | IS-019 | AQELYN Security Data Lake & Telemetry Platform |

The fixed repository structure remains:

```
AQELYN/
■■■ archive/
■■■ blueprint/
■■■ docs/
■■■ src/
■■■ tests/
■■■ tools/
■■■ build/
■■■ releases/
■■■ scripts/
■■■ assets/
■■■ examples/
■■■ plugins/
■■■ sdk/
■■■ api/
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification
↓
Generate Engineering Archive
↓
Continue
```

No Engineering Archive may be skipped.

2. IS-019 Specification Identity

```
Specification ID: IS-019
Name: AQELYN Security Data Lake & Telemetry Platform
Engineering Archive Target: EA-0019
Project: AQELYN
System Type: Cyber Security Operating Environment
Status: Complete
Predecessor: IS-018 - AQELYN Automated Response & Orchestration Engine
```

3. Purpose

The AQELYN Security Data Lake & Telemetry Platform provides the centralized, scalable, and governed storage platform for all security telemetry, logs, events, evidence references, metrics, and operational data generated across the AQELYN ecosystem.

It serves as the authoritative data platform supporting analytics, investigations, threat detection, compliance, digital forensics, machine learning, and long-term retention.

It answers:

```
Where is all AQELYN telemetry stored?
How is security data normalized?
How is long-term retention managed?
How are billions of security events indexed?
How is historical telemetry queried?
How is data integrity preserved?
How are retention policies enforced?
How are analytics supplied with trusted data?
```

4. Mission

The platform shall provide:

```
Centralized telemetry storage
Security Data Lake
High-volume event ingestion
Log normalization
Data indexing
Long-term retention
Immutable storage
Policy-driven retention
High-speed querying
Analytics data services
Evidence linkage
Data governance
```

5. Scope

5.1 In Scope

```
Security telemetry
System logs
Application logs
Event storage
Evidence references
Detection history
Response history
Forensic metadata
Threat intelligence data
Operational metrics
Data indexing
Data lifecycle management
```

5.2 Out of Scope

```
Relational business databases
Enterprise ERP systems
Office document storage
Source code repositories
Media asset management
User productivity platforms
```

6. Dependencies

IS-019 depends on:

```
IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 Identity & Access Governance Engine
IS-012 Asset & Configuration Governance Engine
IS-013 Risk Intelligence Engine
IS-014 Threat Intelligence Fusion Engine
IS-015 Security Operations Engine
IS-016 Digital Forensics Engine
IS-017 Threat Detection & Analytics Engine
IS-018 Automated Response & Orchestration Engine
```

7. High-Level Architecture

```
AQELYN Security Data Lake & Telemetry Platform
■
■■■ Telemetry Ingestion Engine
■■■ Data Lake Engine
■■■ Normalization Engine
■■■ Indexing Engine
■■■ Query Engine
■■■ Retention Engine
■■■ Archive Engine
■■■ Metadata Catalog
■■■ Governance Engine
■■■ Analytics Connector
■■■ Evidence Connector
■■■ Knowledge Graph Connector
■■■ Event Publisher
```

8. Functional Requirements

FR-019-001 - Telemetry Ingestion

The platform shall ingest:

```
Security events
System logs
Application logs
Network telemetry
Identity events
Asset events
Threat intelligence feeds
Response events
```

FR-019-002 - Data Normalization

Normalize:

```
Log formats
Event schemas
Telemetry metadata
Timestamp formats
Source identifiers
Severity levels
```

FR-019-003 - Data Indexing

Support indexing for:

```
Events
Assets
Identities
Threats
Incidents
Evidence
Responses
Investigations
```

FR-019-004 - Data Retention

Support:

```
Retention policies
Legal hold
Archive policies
Immutable storage
Secure deletion
Lifecycle management
```

FR-019-005 - Query Services

Provide:

```
Historical search
Real-time search
Time-range queries
Entity queries
Correlation queries
Analytical queries
```

FR-019-006 - Data Governance

Support:

```
Classification
Labeling
Integrity validation
Retention enforcement
Lineage tracking
Auditability
```

FR-019-007 - Event Publication

Publish standardized events:

```
telemetry.ingested
data.normalized
data.indexed
archive.completed
retention.executed
query.completed
```

9. Non-Functional Requirements

The platform shall provide:

```
Petabyte scalability
High availability
High throughput
Low query latency
Immutable storage
Repository stability
Backward compatibility
Continuous operation
```

10. Core Data Lifecycle

```
Telemetry Received
↓
Normalization
↓
Validation
↓
Indexing
↓
Data Lake Storage
↓
Governance
↓
Retention Management
↓
Archive
↓
Analytics & Query
```

11. Internal Component Architecture

The AQELYN Security Data Lake & Telemetry Platform is implemented as a modular data platform integrated with the AQELYN Kernel, Event Bus, Evidence Engine, Knowledge Graph, Threat Detection Engine, and Automated Response Engine.

```
AQELYN Security Data Lake & Telemetry Platform
■
■■■ Telemetry Ingestion Engine
■■■ Data Lake Engine
■■■ Normalization Engine
■■■ Validation Engine
■■■ Indexing Engine
■■■ Query Engine
■■■ Retention Engine
■■■ Archive Engine
■■■ Metadata Catalog
■■■ Governance Engine
■■■ Analytics Connector
■■■ Evidence Connector
■■■ Knowledge Graph Connector
■■■ Event Publisher
```

12. Component Specifications

12.1 Telemetry Ingestion Engine

Responsible for collecting security telemetry from all AQELYN subsystems.

Capabilities:

```
High-speed ingestion
Streaming telemetry
Batch import
Connector framework
Input validation
```

12.2 Data Lake Engine

Provides centralized storage.

Supports:

```
Immutable storage
Partitioned storage
Distributed storage
Versioned datasets
Long-term retention
```

12.3 Normalization Engine

Normalizes incoming telemetry.

Functions:

```
Schema transformation
Timestamp normalization
Severity normalization
Identity normalization
Metadata enrichment
```

12.4 Validation Engine

Validates incoming data.

Checks:

```
Schema compliance
Integrity
Duplicates
Required metadata
Policy compliance
```

12.5 Indexing Engine

Creates searchable indexes.

Indexes include:

```
Events
Assets
Identities
Threats
Incidents
Evidence
Responses
Investigations
```

12.6 Query Engine

Provides enterprise search.

Supports:

```
Historical search
Time-based queries
Entity queries
Correlation queries
Analytics queries
```

12.7 Retention Engine

Enforces lifecycle rules.

Supports:

```
Retention periods
Legal hold
Archiving
Deletion policies
Compliance retention
```

12.8 Archive Engine

Coordinates archival operations.

Functions:

```
Cold storage
Long-term preservation
Integrity verification
Archive indexing
Recovery support
```

12.9 Metadata Catalog

Maintains dataset metadata.

Stores:

```
Dataset definitions
Lineage
Classification
Ownership
Retention policy
Integrity state
```

12.10 Governance Engine

Enforces governance rules.

Supports:

```
Classification
Integrity validation
Retention enforcement
Audit logging
Data lineage
```

13. Universal Object Model Extensions

13.1 TelemetryRecord

```
TelemetryRecord:
telemetry_id
source
timestamp
classification
integrity
```

13.2 Dataset

```
Dataset:
dataset_id
owner
retention_policy
classification
```

13.3 DataIndex

```
DataIndex:
index_id
entity_type
scope
version
```

13.4 ArchiveRecord

```
ArchiveRecord:
archive_id
retention_state
integrity_hash
archived_at
```

14. Knowledge Graph Integration

Relationships:

```
Telemetry
↓
stored_in
↓
Dataset

Dataset
↓
indexed_by
↓
DataIndex

Evidence
↓
references
↓
Telemetry

Investigation
↓
queries
↓
Dataset

Archive
↓
preserves
↓
Dataset
```

15. Event Bus Integration

15.1 Telemetry Events

```
telemetry.ingested
telemetry.validated
telemetry.rejected
```

15.2 Data Events

```
data.normalized
data.indexed
dataset.created
```

15.3 Archive Events

```
archive.started
archive.completed
retention.executed
```

15.4 Query Events

```
query.started
query.completed
query.failed
```

16. Evidence Engine Integration

Consumes:

```
Evidence metadata
Artifact references
Integrity verification
Evidence lineage
```

17. Knowledge Graph Integration Details

Provides:

```
Entity relationships
Historical references
Threat linkage
Asset linkage
Identity linkage
```

18. Threat Detection Integration

Supplies telemetry to:

```
Behavior analytics
Threat detection
Anomaly detection
Threat scoring
Prediction analytics
```

19. Automated Response Integration

Supplies:

```
Historical telemetry
Response history
Playbook execution history
Recovery history
```

20. Compliance Integration

Supports:

```
Audit datasets
Retention enforcement
Legal hold
Regulatory reporting
Evidence preservation
```

21. Public APIs

21.1 Telemetry API

```
GET /telemetry
POST /telemetry
GET /telemetry/{id}
```

21.2 Dataset API

```
GET /datasets
POST /datasets
```

21.3 Query API

```
POST /query
GET /query/{id}
```

21.4 Archive API

```
GET /archive
POST /archive
```

22. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
■■■ src/
■ ■■■ data_lake/
■■■ tests/
■ ■■■ data_lake/
■■■ docs/
■ ■■■ data_lake/
■■■ api/
■ ■■■ data_lake/
■■■ archive/
```

No top-level repository modifications are permitted.

23. Security Architecture

The AQELYN Security Data Lake & Telemetry Platform is the authoritative storage and telemetry subsystem responsible for securely ingesting, storing, indexing, governing, archiving, and serving security data across the AQELYN platform.

Every stored dataset shall be:

```
Policy-governed
Integrity verified
Immutable (when required)
Fully auditable
Traceable
Versioned
Mission-aware
Risk-aware
```

23.1 Security Principles

```
Zero Trust
Defense in Depth
Least Privilege
Immutable Storage
Data Integrity
Secure by Design
Continuous Monitoring
Governance First
```

23.2 Authorization Model

Supported operational roles:

```
SOC Analyst
Threat Hunter
Digital Forensics Analyst
Data Administrator
Compliance Officer
Mission Owner
Security Administrator
Automation Service
```

All privileged data operations shall be authorized through the AQELYN Policy Engine.

23.3 Data Integrity

Data records shall maintain:

```
Unique telemetry identifier
Source identity
Timestamp
Integrity status
Classification
Lineage
Retention policy
Dataset reference
Audit history
```

Data history shall be append-only where immutability is required.

23.4 Retention Integrity

Retention and archive operations shall maintain:

```
Retention rule reference
Legal hold state
Archive hash
Archive timestamp
Deletion approval
Audit record
Policy reference
```

Retention actions shall be policy-governed and auditable.

24. Data Lifecycle

24.1 Telemetry Lifecycle

```
Telemetry Received
↓
Validation
↓
Normalization
↓
Indexing
↓
```

```
Stored
↓
Queried
↓
Archived
↓
Retention Completed
```

24.2 Dataset Lifecycle

```
Dataset Created
↓
Classified
↓
Indexed
↓
Governed
↓
Archived
↓
Expired
```

24.3 Archive Lifecycle

```
Retention Trigger
↓
Integrity Verification
↓
Archive
↓
Index Update
↓
Verification
```

24.4 Query Lifecycle

```
Query Submitted
↓
Authorization
↓
Index Resolution
↓
Data Retrieval
↓
Audit Recorded
↓
Results Returned
```

25. Continuous Platform Operations

The platform continuously evaluates:

```
Telemetry ingestion health
Storage utilization
Index consistency
Integrity validation
Retention compliance
Archive health
Query performance
Data governance compliance
```

26. Performance Requirements

The platform shall support:

```
Petabyte-scale storage
Millions of events per minute
```

```
Low-latency indexing
Distributed querying
High-speed ingestion
Continuous availability
```

27. Scalability Requirements

The platform shall scale to support:

```
Global deployments
Multi-region clusters
Distributed storage
Hybrid cloud environments
Large enterprise installations
Long-term historical storage
```

28. Audit Requirements

Every platform operation shall generate immutable audit records.

Audit events include:

```
Telemetry ingested
Dataset created
Normalization completed
Index updated
Archive completed
Retention executed
Query executed
Integrity validation
```

29. Failure Handling

29.1 Ingestion Failure

```
Retry initiated
Failure recorded
Alert generated
```

29.2 Storage Failure

```
Replication activated
Integrity checked
Recovery initiated
```

29.3 Index Failure

```
Index rebuilt
Consistency verified
Audit generated
```

29.4 Archive Failure

```
Archive suspended
Retry scheduled
Integrity revalidated
```

30. Testing Strategy

30.1 Unit Testing

Validate:

```
Telemetry Ingestion Engine
Normalization Engine
Validation Engine
Indexing Engine
Retention Engine
Query Engine
```

30.2 Integration Testing

Verify interaction with:

```
Kernel
Event Bus
Evidence Engine
Knowledge Graph
Threat Detection Engine
Automated Response Engine
Policy Engine
Compliance Engine
```

30.3 System Testing

Validate:

```
Telemetry ingestion
Normalization
Indexing
Query performance
Retention
Archiving
```

30.4 Security Testing

Verify:

```
Authorization
Integrity validation
Retention enforcement
Immutable storage
Audit logging
```

30.5 Regression Testing

Verify IS-001 through IS-018 remain unaffected.

31. Acceptance Criteria

IS-019 is complete when:

```
Telemetry Ingestion implemented
Normalization implemented
Indexing implemented
Retention implemented
Archive Engine implemented
Query Engine implemented
Repository unchanged
Testing documented
```

32. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/data_lake/
■■■ tests/data_lake/
■■■ docs/data_lake/
■■■ api/data_lake/
■■■ archive/
```

No top-level repository modifications are permitted.

33. Engineering Summary

IS-019 introduces the AQELYN Security Data Lake & Telemetry Platform, providing enterprise-scale telemetry ingestion, immutable storage, indexing, governance, retention, archival, and analytical query capabilities.

Major capabilities include:

```
Telemetry Ingestion
Security Data Lake
Normalization
Validation
Indexing
Historical Query
Retention Management
Archival
Metadata Catalog
Data Governance
Analytics Services
Evidence Linkage
```

The platform integrates with every previously completed AQELYN engine while preserving repository stability, modularity, and backward compatibility.

34. Specification Status

```
Specification ID : IS-019
Title : AQELYN Security Data Lake & Telemetry Platform
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0019
```

Engineering workflow status:

```
EA-0001 COMPLETE
EA-0002 COMPLETE
EA-0003 COMPLETE
EA-0004 COMPLETE
EA-0005 COMPLETE
EA-0006 COMPLETE
EA-0007 COMPLETE
EA-0008 COMPLETE
EA-0009 COMPLETE
EA-0010 COMPLETE
EA-0011 COMPLETE
EA-0012 COMPLETE
EA-0013 COMPLETE
EA-0014 COMPLETE
EA-0015 COMPLETE
EA-0016 COMPLETE
EA-0017 COMPLETE
```

```
EA-0018 COMPLETE
IS-019 COMPLETE
EA-0019 READY FOR GENERATION
```

35. EA-0019 Engineering Objective

The objective of IS-019 was to introduce a dedicated Security Data Lake & Telemetry Platform that enables AQELYN to ingest, normalize, validate, store, index, govern, archive, retain, and query security telemetry at enterprise scale.

The platform extends AQELYN from response orchestration into data platform capabilities supporting analytics, forensics, detection, compliance, and long-term historical visibility.

36. EA-0019 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

```
Telemetry ingestion
Security data lake storage
Normalization
Validation
Indexing
Query services
Retention enforcement
Archival
Metadata cataloging
Data governance
Analytics connectivity
Evidence connectivity
Knowledge Graph connectivity
Event publishing
```

The platform integrates with all previously completed AQELYN engines while preserving architectural modularity.

37. Major Engineering Decisions

37.1 Decision 1 - Dedicated Data Lake & Telemetry Platform

Telemetry storage and historical query responsibilities are implemented as a standalone platform rather than embedded in Threat Detection, Evidence, or SOC.

Rationale:

```
Clear separation of data platform concerns from analytics and operations.
Independent lifecycle and scaling.
Better support for high-volume telemetry ingestion and long-term retention.
Improved governance over datasets, retention, indexing, and archive policy.
```

37.2 Decision 2 - Evidence References Instead of Evidence Replacement

The Data Lake stores telemetry and references evidence but does not replace the AQELYN Evidence Engine.

Benefits:

```
Preserves Evidence Engine as the authoritative evidence system.
Telemetry can support investigations without duplicating evidence authority.
Forensics and compliance can use both evidence records and telemetry history.
```

37.3 Decision 3 - Governance-First Retention Model

Retention, legal hold, archive, and deletion are governed by policy.

Benefits:

```
Regulatory obligations are enforceable.
Historical telemetry lifecycle is auditable.
Secure deletion and archive decisions are traceable.
Legal hold prevents accidental removal of required data.
```

37.4 Decision 4 - Event-Driven Data Platform

Telemetry, validation, normalization, indexing, archive, retention, and query events are published through the AQELYN Event Bus.

Examples include:

```
telemetry.ingested
telemetry.validated
telemetry.rejected
data.normalized
data.indexed
dataset.created
archive.started
archive.completed
retention.executed
query.completed
```

This maintains loose coupling between AQELYN engines.

37.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

```
TelemetryRecord
Dataset
DataIndex
ArchiveRecord
```

These extend the Universal Object Model without modifying existing object definitions.

38. Architectural Integration Summary

Engine	Integration
IS-001 Kernel	Runtime lifecycle and service registration
IS-002 Universal Object Model	Telemetry, dataset, index, archive objects
IS-003 Event Bus	Telemetry, data, archive, query events
IS-004 Evidence Engine	Evidence metadata, artifact references, lineage
IS-005 Knowledge Graph	Telemetry, dataset, index, evidence, investigation relationships
IS-006 Trust Engine	Integrity state and telemetry trust
IS-007 Mission Engine	Mission-aware retention and analytics context
IS-008 Workflow Engine	Retention, archive, legal hold, governance workflows
IS-009 Policy Engine	Retention, access, archive, classification, legal hold policies
IS-010 Compliance Engine	Audit datasets, retention, legal hold, regulatory reporting
IS-011 Identity Governance Engine	Identity events and access telemetry

| IS-012 Asset Governance Engine | Asset events and configuration telemetry |

| IS-013 Risk Intelligence Engine | Historical risk data and risk analytics |

| IS-014 Threat Intelligence Engine | Threat intelligence data and indicators |

| IS-015 SOC Engine | Incident, case, and investigation telemetry |

| IS-016 Digital Forensics Engine | Forensic metadata, artifacts, reports |

| IS-017 Threat Detection Engine | Detection telemetry, behavior analytics, anomaly data |

| IS-018 Response Orchestration Engine | Response, playbook, recovery telemetry |

No existing engine required redesign.

39. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/data_lake/
■■■ tests/data_lake/
■■■ api/data_lake/
■■■ docs/data_lake/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

40. Security Impact Summary

The specification introduces data-lake-specific security controls:

```
Policy-driven data access
Immutable storage where required
Integrity verification
Retention enforcement
Legal hold
Archive integrity validation
Audit trail for queries and lifecycle actions
Dataset classification
Lineage tracking
Governed secure deletion
```

No reduction in the security posture of existing components was identified.

41. Capabilities Added

The platform enables AQELYN to support:

```
Centralized telemetry storage
Security Data Lake
High-volume event ingestion
Log normalization
Data validation
Data indexing
Long-term retention
Immutable storage
Policy-driven retention
High-speed querying
Analytics data services
Evidence linkage
Metadata cataloging
```

```
Data governance
Archive management
```

42. Risks Identified

Risk	Mitigation
Excessive data volume	Distributed storage, partitioning, indexing, retention
Query performance degradation	Indexing engine and query lifecycle management
Improper retention	Policy-governed retention and legal hold
Data integrity loss	Integrity validation and immutable storage
Unauthorized data access	Policy enforcement and role authorization
Archive failure	Retry, integrity revalidation, audit
Data normalization errors	Validation engine and schema compliance
Evidence authority confusion	Evidence Engine remains source of truth

No critical architectural risks were identified that require redesign.

43. Verification Summary

The specification defines verification for:

```
Unit testing
Integration testing
System testing
Security testing
Regression testing
```

Acceptance criteria cover telemetry ingestion, normalization, indexing, retention, archive engine, query engine, repository validation, and testing documentation.

44. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

```
Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline
```

45. Dependencies

Required:

```
EA-0001 through EA-0018
IS-001 through IS-018
```

Enables:

```
IS-020 and subsequent AI, analytics, reporting, decision intelligence, telemetry, and platform-scale
components
```

46. Completion Record

```
Engineering Archive : EA-0019
Implementation Specification : IS-019
Title : AQELYN Security Data Lake & Telemetry Platform
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue
```

47. Archive Index Update

```
EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations (SOC) Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine
EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
EA-0019 IS-019 AQELYN Security Data Lake & Telemetry Platform
```

48. Engineering Phase Status

```
Completed Engineering Archives : EA-0001 through EA-0019

Current Status:
EA-0019 COMPLETE

Next Implementation Specification:
IS-020 - AQELYN AI Decision Intelligence Engine
```

EA-0019 is completed and archived. The engineering workflow is consistent with the project rule:

```
Implementation Specification -> Engineering Archive -> Continue
```

From this point onward, the next engineering artifact is IS-020.

49. Engineering Archive Publication Standard

EA-0019 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
■
■■■ diagrams/
■ ■■■ Architecture.svg
■ ■■■ Component.svg
■ ■■■ Workflow.svg
■ ■■■ EventFlow.svg
■ ■■■ Integration.svg
■
■■■ examples/
■ ■■■ example_*.md
■
■■■ HTML/
■ ■■■ index.html
■ ■■■ styles.css
■ ■■■ assets/
■
■■■ images/
■
■■■ journal/
■ ■■■ Engineering_Journal.md
■
■■■ manifest/
■ ■■■ manifest.json
■
■■■ MD/
■ ■■■ EA-xxxx.md
■
■■■ PDF/
■ ■■■ EA-xxxx.pdf
■
■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

50. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-019-001	Ingest telemetry	Sections 8, 12	Complete
FR-019-002	Normalize data	Sections 8, 12, 24	Complete
FR-019-003	Index data	Sections 8, 12, 24	Complete
FR-019-004	Support retention	Sections 8, 12, 23, 24	Complete
FR-019-005	Provide query services	Sections 8, 12, 24	Complete
FR-019-006	Support data governance	Sections 8, 12, 23	Complete
FR-019-007	Publish data events	Sections 8, 15, 37	Complete
NFR-019-001	Petabyte scalability	Sections 9, 26, 27	Complete
NFR-019-002	High availability	Sections 9, 26	Complete
NFR-019-003	High throughput	Sections 9, 26	Complete
NFR-019-004	Low query latency	Sections 9, 26	Complete
NFR-019-005	Immutable storage	Sections 9, 23, 40	Complete
NFR-019-006	Repository stability	Sections 22, 32, 39	Complete

51. Traceability Matrix

Source	Target	Relationship
IS-019 Purpose	EA-0019 Objective	Defines why the platform exists
Telemetry Ingestion Engine	FR-019-001	Implements data ingestion
Normalization Engine	FR-019-002	Implements schema and metadata normalization
Indexing Engine	FR-019-003	Implements searchable indexes
Retention Engine	FR-019-004	Implements retention and legal hold
Query Engine	FR-019-005	Implements search and analytical queries
Governance Engine	FR-019-006	Implements classification, lineage, and retention enforcement
Event Publisher	FR-019-007	Publishes telemetry, data, archive, and query events
Evidence Engine Integration	Evidence linkage	References evidence metadata and lineage
Knowledge Graph Integration	Data relationships	Links telemetry, datasets, evidence, investigations
Threat Detection Integration	IS-017	Supplies analytics telemetry
Automated Response Integration	IS-018	Supplies response history
Compliance Integration	IS-010	Supports retention, audit datasets, and regulatory reporting
Repository Validation	Repository Standard	Confirms no top-level redesign

52. Engineering Journal

Journal Entry - EA-0019

EA-0019 was created to archive completion of IS-019 - AQELYN Security Data Lake & Telemetry Platform.

The archive records the expansion of AQELYN into enterprise-scale telemetry and data platform capabilities. IS-019 defines the structure needed to ingest telemetry, normalize logs and events, validate data, store data in a governed data lake, create indexes, support query services, enforce retention, manage archives, catalog metadata, govern datasets, and serve analytics consumers.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Telemetry storage must be modeled separately from evidence, threat detection, and SOC operations. The Evidence Engine owns authoritative evidence records, the Threat Detection Engine consumes telemetry for analytics, SOC consumes operational history, and the Data Lake & Telemetry Platform owns high-volume security data ingestion, storage, indexing, retention, archive, and query services.

Governance Note

EA-0019 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

53. Examples

53.1 Example Telemetry Record

```
telemetry_id: TEL-0001
source: aqelyn_event_bus
timestamp: 2026-07-07T12:00:00Z
classification: security_event
integrity:
status: verified
dataset: DATASET-SECURITY-EVENTS
```

53.2 Example Dataset

```
dataset_id: DATASET-SECURITY-EVENTS
owner: data_administrator
retention_policy: RETENTION-SECURITY-365D
classification: security_sensitive
integrity: verified
```

53.3 Example Archive Record

```
archive_id: ARCH-1001
retention_state: archived
integrity_hash: sha256:7a0f-example
archived_at: 2026-07-07T12:30:00Z
dataset: DATASET-SECURITY-EVENTS
```

53.4 Example Data Event

```
{
"event_type": "telemetry.ingested",
"telemetry_id": "TEL-0001",
"dataset_id": "DATASET-SECURITY-EVENTS",
"source_engine": "aqelyn_security_data_lake_telemetry_platform"
}
```

54. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0019.md
PDF/EA-0019.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_data_lake.md
```

55. Final Archive Statement

EA-0019 is the Engineering Archive for IS-019 - AQELYN Security Data Lake & Telemetry Platform.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0019 COMPLETE
IS-019 COMPLETE
NEXT: IS-020
```